Q: = 25% 200

$\frac{25}{100} \times 200 = 50$

b. 50

Q2. 40% of 80

~~$\frac{40}{100} \times 80 = 32$~~

$\frac{40}{100} \times x = 80$

$x = \frac{80}{4/10}$

$x = \frac{80}{0.4}$

$x = 200$

C = 200

Q3 = 75% of num is 150

$$75\% = \frac{75}{100} \times x = 150$$

$$x = \frac{150}{\frac{75}{100}}$$

$$x = 200$$

$$\underline{b = 200}$$

Q4 = 15% of 120

$$\frac{15}{100} \times 120 = 18$$

$$\underline{c) = 18}$$

Q5 30% of num is 90

$$\frac{30}{100} \times x = 90$$

$$x = \frac{90}{0.30}$$

$$x = 300$$

$$c = 300$$

Q6. price of product increase ₹200 to 250

$I = 2500 - 200$
$= 50$

increase $= \frac{50}{200}$
$= 0.25$

$P = 0.25 \times 100$
$= 25\%$

Q7 Salary incr 40,000 to 50000

$I = 50000 - 40000$
$= 10000$

Increase $= \frac{10000}{40000}$

$= \frac{1}{4} = 0.25$

$P = 0.25 \times 100$
$= 25\%$

Q8. decrease $= 10000 - 8000$

$= 2000$

$d = \frac{2000}{10000}$

$= \frac{2}{10} = 0.2$

$P = 0.2 \times 100$

$= \underline{\underline{20\%}}$

Q9 price drop $= 500 - 400$

$= 100$

$d = \frac{100}{500}$

$= \frac{1}{5} = 0.2$

$P = 0.2 \times 100$

$= \underline{\underline{20\%}}$

Q10 $CP = 600$ $SP = 450$

$d = 600 - 450$

$= 150$

$D = \frac{150}{600}$

$= 1/4 = 0.25$

$P = 0.25 \times 100$

$= \underline{\underline{25\%}}$

Q11. 30% of 400

$= \left(\frac{30}{100}\right) \times 400$

$= 120$

40% of 300

$= \left(\frac{40}{100}\right) \times 300$

$= 0.40 \times 300$

$= 120$

Both are equal

Q12 Saving percentage = 100% − 60% = 40%

40% = 8000

$= \left(\frac{40}{100}\right) \times x = 8000$

$= 0.40 \times x = 8000$

$= x = \frac{8000}{0.40}$

$x = \underline{\underline{20000}}$

Q13 A is 20% more than B

let B = 100 A = 120

D = 120 − 100

= 20

$Pl = \left(\frac{20}{120}\right) \times 100$

$= \underline{\underline{16.67}}$

Q14

Q

Q14

Percentag Reduce $= \left(\frac{25}{(100+25)}\right) \times 100$

$= \left(\frac{25}{125}\right) \times 100$

$= \boxed{20\%}$

Q15 40% more than B

$PL = \left(\frac{40}{100+40}\right) \times 100$

$= \frac{40}{140} \times 100$

$= \underline{\underline{28.57\%}}$

Q16. original = 100

increase by 20% : $100 + \left(\frac{20}{100}\right) \times 100$

$= 120$

increase by 10% $= 120 - \left(\frac{10}{100}\right) \times 120$

$= 120 - 12$

$= 108$

Net Change $= 108 - 100$

$= 8$

$\underline{\underline{8\%}}$ increase

Q17. let $N = 100$

incr 30% $= 100 + \left(\frac{30}{100}\right) \times 100$

$= 130$

incr 20% $= 130 - \left(\frac{20}{100}\right) \times 130$

$= 130 - 26$

$= 104$

Net change $= 104 - 100$

$= 4$

4% increase

Q18. let p 100

Incr 25% $= 100 + \left(\frac{25}{100}\right) \times 100$

$= 125$

Incr 20% $= 125 - \left(\frac{20}{100}\right) \times 125$

$= 100$

Net change $= 100 - 100$

$= 0$

0%

Q19. let $P = 100$

Incr 40% $= 100 + \left(\frac{40}{100}\right) \times 100$

$= 140$

Incr by 30% $= 140 - \left(\frac{30}{100}\right) \times 140$

$= 140 - 42$

$= 98$

Net change : $98 - 100$

$= 2$

Net change = <u>2% decrease</u>

Q20. let $S = 100$

Incr 20% : $100 + \left(\frac{20}{100}\right) \times 100$

$= 120$

Incr 10% $= 120 - \left(\frac{10}{100}\right) \times 12$

$= 120 - 12$

$= 108$

Net change $= 108 - 100$

$= 8$

<u>8% increase</u>

Q21 CP = 100%

Profit : 25%

SP = 100% + 25%

= 125%

SP = <u>125%</u>

Q22

$SP = MP - Dis$

Discount $= 10\%$ of 500

$= \left(\frac{10}{100}\right) \times 500$

$= 50$

$SP = 500 - 50$

$= 450$

Cost price $= 450 = x + \left(\frac{8}{100}\right) \times x$

$450 = x\,(1 + 0.08)$

$450 = 1.08x$

$x = \frac{450}{1.08}$

$= ₹\ 416.67$

$= \underline{\underline{420}}$

Q23 let CP be 100

Profit $= 20\%$ of 100

$= 20$

$SP = CP + \text{profit} = 100 + 20$

$= 120$

$PP = \left(\frac{20}{120}\right) \times 100$

$= \underline{\underline{16.67}}$

Q24

Discount = 1200 − 960

= 240

Percentage Dis = $\left(\frac{240}{1200}\right) \times 100$

= 20%

Q25). Profit = 650 − 500

= 150

Per Profit = $\left(\frac{150}{500}\right) \times 100$

= 30 %

Q26 let B = 100 A = 120

120 − 100

= 20

Pl = $\frac{20}{120} \times 100$

= 16.67

Q27 ratio = 3 + 2 = 5

Percentage boys = $(3/5) \times 100$

= 60%

Q28 Incr = 2500000 − 200000

= 50,000

Perc Incr = $\left(\frac{50000}{2500000}\right) \times 100$

= 25 %

Q29 = Cand A = 65% B = 35%

Diff = 65 − 35
= 30%

Vote 30% = 3000

Total Vote = $\frac{3000}{0.30}$
= 10000

Q30. Percentage increase = $\left(\frac{30}{100-30}\right) \times 100$

$= \frac{30}{70} \times 100$

= 42.85%

Q31 let N = 100

incr 50% of 100 = 150

~~incr 50~~ decr by 50% = $150 - \frac{50}{100} \times 150$

= 75

Net change = 75 − 100
= 25% decrease

Q32

Q33

Q34

Q32 A 20% taller than B

$120 - 100$

$= 120$

$Pl = \frac{20}{120} \times 100$

B is shorter than A $= \underline{16.67\%}$

Q33 If 30% is 90

$10\% = \frac{90}{3}$

$= 30$

Therefore $= 60\%$ is 30×6

$= 180$

$\underline{180}$

Q34 spend 75%

$= 100\% - 75\%$

$= 25\%$

25% of income $= 5000$

Total $= \frac{5000}{(25/100)} = 5000 \times 4 = 20000$

$\underline{₹20000}$

Q35. 20% increase

let $120 - 100 = 20\%$

Consumption reduce $= \frac{20}{120} \times 100$

$= 16.67\%$

$\underline{\underline{16.67\%}}$

Q36 Intial price $= 100 + (20)$

$= 120$

incr by 20% $= 100 + 20$

$= 120$

decr by 10 $= 120 - 12$

$= 108$

Net change $= 108 - 100$

$= 8$

Percent Change $= \left(\frac{8}{100}\right) \times 100$

$= 8\%$ increase

Q37. let CP $= 100$

MP $= 100 + 25 = 125$

Discount 20% $= \left(\frac{20}{100}\right) \times 125$

$= 25$

SP $= 125 - 25 = 100$

Profit $= 100 - 100$

$= \underline{\underline{0}}$

Q38 CP = 500 loss = 20%

$loss = \frac{20}{100} \times 500$

$= 100$

$SP = 500 - 100$

$= 400$

Q39 let sal 100

incr by 10% $= 100 + 10$

$= 110$

Decr by 10% $= 110 - \left(\frac{10}{100}\right) \times 110$

$= 110 - 11$

$= 99$

Net change $= 1$

1% decrease

Q40 Passing No 40%

Passing marks $= 200 + 20$

$= 220$

40% of total marks $= 220$

$\frac{220}{(40/100)} = 220 \times \left(\frac{100}{40}\right)$

$= 550$

Q41. Total per $= 20 + 30 + 10$

$= 60\%$

Saving per $= 100 - 60$

$= 40$

40% of Salary $= 18000$

Total Sal $= \dfrac{18000}{\left(\frac{40}{100}\right)}$

$= \underline{\underline{45000}}$

Q42 let cost be 100

Incr 30% $= 100 + 30$

$= 130$

Decr 30% $= 130 - \left(\frac{30}{100}\right) \times 130$

$= 130 - 39$

$= 91$

Net change $: 91 - 100$

$= 9$

Net percentge change $= \underline{\underline{9\%}}$ $\underline{\underline{\text{decrease}}}$

Q43. 1 year $= 10000 \times 1.10$

$= 11000$

2nd year $= 11000 \times 1.10$

$= 12100$

3rd year $= 12100 \times 1.10 =$

$= \underline{\underline{13310}}$

Q44 15% of A = 20% of B

$0.15A = 0.20B$

$A/B = \frac{0.20}{0.15} = \frac{20}{15} = 4/3$

$A:B = \underline{\underline{4:3}}$

Q45 Profit $= \left(\frac{25}{100}\right) \times 800$

$= 200$

$SP = 800 + 200$

$= \underline{\underline{1000}}$

Q46. Profit $= 250 - 200$

$= 50$

Profit percentage $= \left(\frac{50}{200}\right) \times 100$

$= \underline{\underline{25\%}}$

Q47 $CP = x$

$1.20x = 720$

$x = \frac{720}{1.20}$

$= \underline{\underline{600}}$